How to push Elmina Canopy Prioritiser closer to a winning project

After reviewing your latest repository, SDP's actual judging rubric, Sime Darby Property's biodiversity strategy and research on urban tree planning, I see five strong opportunities beyond the UX redesign and Validate–Review–Replan workflow.

The biggest opportunity is to make your system more specifically useful to Sime Darby Property, rather than more technically complicated.

Your current project already has satellite analysis, multi-objective rankings, impact reach, planting plans, conflict flags, resident-exposure scenarios and an OSM-only fallback.



GitHub

Adding another generic dashboard metric probably won't move the judging score much.

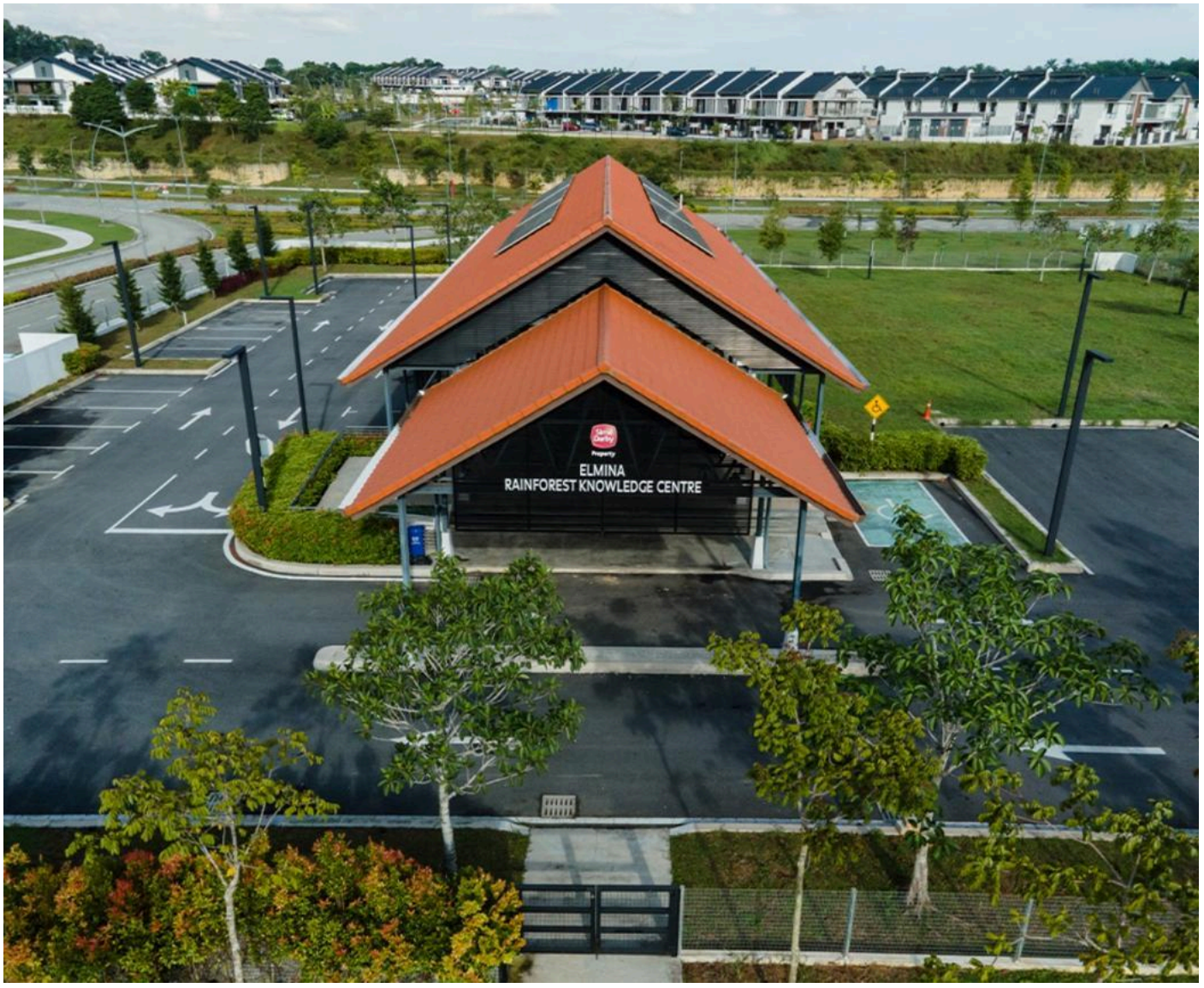
The five improvements I would prioritize

Priority	Improvement	Main SDP criterion	Potential value
1	SDP biodiversity and conservation planner	Strategic Fit 15%, Impact 25%	Very high
2	Resource-constrained planting optimization	Innovation 20%, Commercial 10%	Very high
3	Walking-route shade gap analysis	Impact 25%, Innovation 20%	High
4	Tree establishment and survival planning	Impact 25%, Feasibility 20%	High
5	Real second-township demonstration	Commercial 10%, Feasibility 20%	High, but costly

These are priorities, not promised score increases. They're designed to address weaknesses in the official rubric, which emphasizes measurable sustainability value, originality, practicality and scale.

Sime Darby Property University Hackathon

1. Make it an SDP biodiversity planner, not just a tree-placement tool



This is probably your strongest problem-fit opportunity.

SDP describes its biodiversity approach through three principles: Avoid, Regenerate and Conserve. Its Elmina programme also focuses on native trees, ecological corridors and biodiversity restoration.

Sime Darby Property

Your current system partly addresses this with its corridor-connectivity score, but it's still centred on ranking places to plant more trees.

A better decision-support system would ask a question first:

Does this location need new trees, restoration, protection of existing vegetation, or further assessment?

Add an intervention recommendation

Intervention	What the planner sees
Protect existing vegetation	Preserve valuable existing trees or habitat; field review required
Regenerate habitat	Consider restoration near corridor gaps with ecologist input
Plant for pedestrian shade	Identify candidate areas near poorly shaded routes
Inspect before deciding	Environmental priority is known, but suitability is unresolved

This creates a stronger distinction between environmental priority and the appropriate ecological intervention.

Important: Your satellite vegetation proxy cannot identify tree maturity, native species or habitat quality reliably. The system should ask for verified observations rather than automatically prescribing conservation actions.

Research and biodiversity guidance support considering multiple ecosystem benefits and existing vegetation, not simply maximizing tree counts.



Forest Research

Implementation: Add an intervention-type field, supporting reasons and expert-verification requirements to the existing plan. Start with manually reviewed classifications.

2. Optimize the best planting combination, not just the highest-ranked individual sites

This may be your most promising technical differentiator.

Your current system ranks individual sites and already supports a spread-out shortlist and deduplicated reach. The next step is to make those concepts part of an actual resource-constrained selection process.

Suppose SDP has resources to inspect only 10 candidates this month.

Simply taking the top 10 scores may not give the best combined coverage. Several high-ranking candidates could serve overlapping footpaths or the same homes.

A planner would benefit from asking:

“Given our inspection capacity, which combination of candidate sites covers the greatest number of distinct opportunities?”

Illustrative optimization

CONCEPTUAL PILOT OPTIMIZER

What are we prioritizing?
Unshaded footpath reach
Unique residential buildings near candidates
Approximate biodiversity corridor gaps
Number of candidates to inspect

10

5
25
50

Selection objective

Choose candidates with the greatest additional unique footpath reach.

Exclude rejected locations, preserve unknown constraints, and compare with ordinary top-10 ranking.

Design this optimizer Concept only: the selector above does not run optimization or display calculated results.

A published multi-objective urban tree-planning framework considers real resource limits and different environmental goals, which supports the relevance of this approach. Optimization itself is not novel; your differentiation would be connecting it to SDP's actual decision requirements and Elmina's evidence.



ScienceDirect

For your repo, I'd implement a greedy marginal-reach selector first, compare it against the current Ranked and Spread out approaches, and keep the original ranking intact.

Do not call the resulting choice environmentally optimal in the real world: it would be optimal or near-optimal only against the selected proxy objective and assumptions.

3. Change “near unshaded footpaths” into meaningful shade-route opportunities




UMMY WAX
SERI WAJAH
NISMA NORIS



10

Currently, your walking-route assessment focuses on OSM footpath and vegetation information near proposed sites.

That helps with screening, but a resident doesn't necessarily walk on every footpath near home. What would be more useful is to identify important route connections.

For example, a planner might investigate:

- Which mapped walking routes connect residential areas to parks, schools or transit stops?
- Which segments have low vegetation-cover proxies?
- Where could a proposed intervention address a gap along an otherwise greener route?

A study on street-tree optimization illustrates the trade-off between providing shade and placing it where people are likely to benefit.

Penn Design

My implementation preference: Create a route-gap screening layer from the OSM walking network, with selectable origins and destinations. Show disconnected routes and missing map data explicitly.

This strengthens the People co-benefit while retaining Planet as your competition track.

However, don't label the result “walking demand” without observed pedestrian counts, and don't treat NDVI as measured shade at a particular hour.

4. Add establishment and maintenance readiness

Research on urban tree mortality consistently finds that species, site conditions and stewardship affect survival. A US Forest Service literature review found particularly important mortality risks in the early years after planting.



US Forest Service Research and Development

Your product would be more useful if its proposed pilot plan included who is responsible for keeping the trees alive.

Add a small maintenance-readiness section to each inspected planting proposal:

Field	Example of permitted input
Species	Awaiting ecologist selection
Soil/drainage	Not checked
Watering availability	Unknown
Maintenance responsibility	Unassigned
Establishment inspection schedule	To be set by planner
Survival observation	No field data

This is more useful than generating speculative survival percentages.

A future real pilot could then collect survival and maintenance records, giving the project a credible pathway toward measuring actual impact.

5. Demonstrate replication beyond Elmina

You already built an independent OSM-only fallback, but your commercial documentation still acknowledges that another township requires adaptation and has not been tested.



GitHub

One genuine second-township run would be more convincing than further theoretical scalability claims.

I would make the OSM-only pipeline accept a different area boundary and township name, generate candidates and exports in an isolated output folder, and run the data self-check.

That proves geographic portability of the OSM-only screening workflow, not equivalence with the satellite model or readiness for all SDP sites.

Given the effort required, this belongs after the higher-priority improvements.

What would actually improve winning potential?

There are two distinctions I would make.

First, features can improve the plausibility of an impact claim, but independent evidence improves its credibility. Your work can become more sophisticated without proving that a tree will survive, that pedestrians will receive extra shade or that SDP can use the result.

Second, strategic fit is especially valuable when it is company-specific. Your project should not look like a canopy dashboard adapted to any random city. It should clearly support Elmina's actual biodiversity-restoration and township-management priorities.

Research from 2025 on urban planting decisions also shows the importance of considering planting potential, heat benefits, implementation difficulty and equity together. That reinforces your direction, but it means your strongest differentiation is the integrated, usable local workflow—not simply using multiple scores.



ScienceDirect

Recommended development order

I would split the remaining work into two horizons, because the proposal deadline is 23 October 2026, while successful Top 10 teams have a later mentoring and development round.

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Development	**Before 23 Oct**	**If selected for Top 10**

SDP Avoid–Regenerate–Conserve intervention review	Build small version	Validate with ecologist
Constrained shortlist optimizer	Build if tests can be completed	Add realistic resource constraints
Route-gap analysis	Research/prototype only	Implement and validate routes
Maintenance readiness	Add planning fields	Collect actual planting records
Second-township proof	Defer	Run independent geographic test
Independent user or field evidence	Do now if practical	Expand and repeat

The first two are my preferred additional software features. They build on components you already have and offer meaningful differentiation.

But there is one non-software activity that could outperform all five in judging value: have an independent landscape professional evaluate the workflow and record their feedback. A brief, genuine review could identify missing planting constraints and validate whether the decision process makes operational sense. Don't claim SDP endorsement unless it comes from an authorized SDP representative.

My final recommendation

I would develop the product around this distinctive concept:

Elmina Canopy Prioritiser — from environmental screening to evidence-based biodiversity and planting decisions.

It would help SDP answer three questions:

1. What does this location actually need? Protection, regeneration, planting or further investigation.
2. Which combination of actions offers the best potential value within available resources? A constraint-aware shortlist, not just individual scores.
3. How will we verify that the decision was appropriate? Field observations, expert review and eventual maintenance outcomes.

That is a stronger problem-solution fit than adding another map visualization.

I would not promise that these features raise a provisional 75 to 85 or 90. The missing real-world evidence still matters enormously. But company-specific conservation decisions plus constrained resource allocation

are the two additions I would choose for greater innovation, practical relevance and potential to stand out to SDP judges.